

section 5 below. Overall, the very high risk aversion during bad times dominates the low risk aversion during good times, resulting in a high long-run equity risk premium.

If time-varying risk aversion, which increases dramatically during bad times, is responsible for the equity risk premium, then investors should ask the following question before holding large equity stakes: In bad times, say, of low consumption growth in the basic Mehra–Prescott model or bad times more generally in a multifactor context, do you behave in a less risk-averse fashion than the market?

Most investors and the representative investor, by construction, become much more risk averse during recessions: bad times are spooky, so everyone gets scared—but to greater or lesser degrees. The question is whether you can tolerate these bad times better than the average investor. The Norwegian sovereign wealth fund certainly can since it has stable cash flows, no immediate liabilities, and a well-functioning governance structure. Supreme Court Justices are well situated to tolerate bad times since their income stream is risk free (unless they have borrowed money to the hilt). And many private wealth management clients also find bad times for the economy hurt less for them than for the average investor. These special types of investor have lower risk aversions during downside events and can tolerate larger losses during bad times. They can hold more equities. In doing so, they will earn higher average returns.

3.2. DISASTER RISK

Disaster models take bad times to the extreme. The bad times are really, really bad: they represent catastrophic declines in consumption. In this explanation, the equity premium is the reward to compensate for rare catastrophes.⁴ This story was introduced by Rietz (1988) and immediately dismissed by Mehra and Prescott (1988) as lacking empirical evidence. But the disaster explanation has made a recent roaring comeback beginning with Barro (2008), who examined the data much more thoroughly. Catastrophes are labeled “black swans” by Nassim Taleb (2004) in the popular literature, although he would object to having them treated in a rational paradigm.⁵ They cannot be predicted or at least are very hard to predict, and for disasters to account for high equity risk premiums, the disasters must be very bad indeed.

⁴More generally, they do not need to be large, sudden negative jumps per se. They can be a series of persistent, large losses in the left-hand tails. Martin (2012) shows that asset prices reflect the possibility of extraordinary bad news that nests but does not necessarily need the assumption of aggregate disaster risk.

⁵The recent financial crisis cannot be considered as an example of an averted disaster to explain the equity premium. While there was a downturn in economic growth and consumption, the downturn happened as a result of the crash in financial prices, rather than the other way round. In disaster models, sharp contractions of consumption happen contemporaneously with falls in asset prices.